

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7zZWmnRuKu4KfuUJtvSb7H4uH8PUDq26QK8QY9TG81LJ

Date: 4/30/2025

Compounds: Canabis, Barbiturates

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Canabarbital

Formula: C₂₁H₂₆N₂O₃

Weight: 354.44

Drug-likeness: 0.71

Activity:

Sedative, anxiolytic

Target Analysis

GABA_A receptor

Binding Affinity: -7.5

Primary target for sedation and anti-anxiety effects.

CB1 cannabinoid receptor

Binding Affinity: -6.8

Modulation of psychoactive effects and appetite stimulation.

Toxicity Profile

Risk Level: High

Affected Systems: Liver, Respiratory system, Central nervous system

Mitigation Suggestions:

- Dose monitoring
- Regular liver function tests
- Avoid use in patients with respiratory disorders

Pharmacokinetics

Absorption:

Rapid

Distribution:

Widely distributed, crosses blood-brain barrier

Metabolism:

Hepatic

Excretion:

Renal

Half-life: 10-30 hours

Detailed Analysis

The new compound, Canabarbit, represents a combination of active moieties from barbiturates and cannabinoids, intended to leverage the sedative and anxiolytic characteristics of both classes of drugs. The primary mechanism involves dual modulation of GABA_A and CB1 receptors, potentially leading to enhanced inhibitory effects on the central nervous system. While promising for certain therapeutic applications, the compound carries significant toxicity risks, particularly in terms of central nervous system depression and hepatotoxicity, necessitating careful monitoring and tailored administration protocols. Through this innovative approach, Canabarbit contributes to the field of anxiolytics with its multifaceted mode of action, but comprehensive clinical validation will be essential to ensure safety and efficacy.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>